
When Does Simulated Quantum Machine Learning Beat Classical Machine Learning? A 20-Qubit Study with Honest Scope

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Abstract

Quantum machine learning (QML) is widely promoted as a route to learning tasks that classical machine learning cannot handle, yet the field’s central question—*when does a quantum model actually beat the best classical model?*—remains unsettled and frequently overstated. The honest answer is two-sided: rigorous classical-hard/quantum-easy separations exist but require fault-tolerant hardware and are not simulable at 20–30 qubits, while the only demonstrable separations at simulable scale live on engineered tasks. We provide a single, reproducible study that pins down both sides at *20 qubits or more* under exact statevector simulation. On an engineered task built along the direction of maximal geometric difference between a quantum and a classical kernel, a projected-quantum-kernel SVM reaches 0.865 test accuracy at 20 qubits while the best of six tuned classical models reaches 0.629 (paired t -test $p = 3.9 \times 10^{-8}$, Cohen’s $d = 5.4$); the advantage is stable at ≈ 0.22 – 0.23 across 8, 10, and 20 qubits. A within-quantum control shows the naive fidelity kernel collapses to chance (0.493) at 20 qubits, confirming the win comes from the *right* quantum inductive bias rather than quantumness alone. On the discrete-logarithm task, six off-the-shelf classical models stay at chance (0.49–0.52) for 12/17/21-bit problems while the quantum interval-state kernel separates the task near-perfectly (0.997–1.000). We document every caveat—engineered labels, asymptotic hardness, and noiseless infinite-shot simulation—so the claim is confirmed with honest scope: a 20+-qubit simulated quantum kernel generalizes on tasks where the best classical models cannot.

1 Introduction

Quantum machine learning (QML) is often presented as a path to solving learning problems that classical methods cannot. But the claim that “quantum beats classical” is rarely accompanied by the only thing that makes it scientific: a concrete task, a strong classical baseline that demonstrably fails, and a scale at which the comparison is actually run. The field’s central question is sharp and still open—*when, if ever, does a quantum model beat the best classical model on a learning task?* Answering it honestly matters: it protects practitioners from hype and points research at the regimes where quantum methods can genuinely help.

The literature gives a two-sided answer. On one hand, there are *provable* classical-hard/quantum-easy separations: the discrete-logarithm construction of Liu et al. [2021] shows that, under the classical hardness of the discrete log, no efficient classical learner can beat random guessing while a quantum support vector machine achieves near-perfect accuracy. On the other hand, these separations embed Shor’s algorithm and live in the fault-tolerant, hundreds-of- qubits regime; at any classically simulable scale (≤ 30 qubits) the underlying number-theoretic problem is trivially solvable, so the separation *cannot* be exhibited by a 20-qubit simulation. The only separations that *are* demonstrable at simulable scale—Huang et al. [2021] and Jerbi et al. [2023]—are on *engineered*

data, relabeled so that the labels align with a quantum kernel. Meanwhile, cautionary results show that expressivity alone destroys generalization [Kübler et al., 2021], every quantum model is a kernel method [Schuld, 2021], and many flagship speedups dequantize [Tang, 2019].

The gap. Despite this rich theory, a clean experimental anchor is missing. Published demonstrations are either small-scale (≤ 12 qubits, e.g., Kübler et al., 2021, Jerbi et al., 2023), run on hardware with no defeated classical baseline (≤ 17 qubits [Peters et al., 2021], 27 qubits [Glick et al., 2024]), or purely theoretical [Liu et al., 2021, Schuld, 2021]. A single, reproducible ≥ 20 -qubit simulation that pits a quantum kernel against a *full* classical suite, reports the geometric-difference diagnostic g before training, quantifies scaling and statistical significance, and states the engineered-task caveat explicitly, has not been collected in one place.

Our study. We provide exactly that. We ask whether there are tasks where classical ML struggles while quantum methods—simulated at 20 qubits or more—excel, and we answer it in two complementary experiments, both reproduced from published constructions and run under exact statevector simulation in PennyLane `lightning.qubit`. Experiment A reproduces the projected-quantum-kernel construction of Huang et al. [2021]: inputs are encoded into a 20-qubit state, labels are engineered along the direction of maximal geometric difference between the quantum and classical kernels, and we compare the quantum-kernel SVM against six tuned classical models over 10 repeated stratified splits. Experiment B uses the discrete-logarithm task of Liu et al. [2021]: six off-the-shelf classical models are trained on bit features while the quantum interval-state kernel is computed from the (simulable) log table.

Preview of results. On the engineered task at 20 qubits, the projected quantum kernel (PROJECTED-QK) reaches 0.865 test accuracy while the best classical model (gradient boosting) reaches 0.629—a gap of +0.236 with a highly significant paired test ($p = 3.9 \times 10^{-8}$, $d = 5.4$). The advantage is *stable* at ≈ 0.22 –0.23 across 8, 10, and 20 qubits. A within-quantum control is decisive: the naive fidelity kernel (FIDELITY-QK) collapses to chance (0.493) at 20 qubits—exactly the high-dimensional concentration the theory predicts—so the win comes from the *right* kernel, not from quantumness. On the discrete-log task, all six classical models sit at chance (0.49–0.52) across 12/17/21-bit problems while INTERVAL-QK separates the task near-perfectly (0.997–1.000).

Contributions.

- We deliver the first self-contained, reproducible ≥ 20 -qubit simulation study that pits a quantum kernel against a *full* six-model classical suite, reports the g diagnostic before training, and quantifies scaling and paired significance ($p \approx 4 \times 10^{-8}$, $d = 5.4$ at 20 qubits).
- We establish a within-quantum control showing the fidelity kernel concentrates to chance at 20 qubits while the projected kernel wins, isolating the advantage to a problem-matched quantum inductive bias rather than quantumness per se.
- We demonstrate off-the-shelf classical failure on the discrete-log task across 12/17/21 bits, where tree models reach perfect train accuracy yet generalize at chance—the signature of a label that is random in the input representation.
- We document every caveat (engineered labels, asymptotic hardness, noiseless infinite-shot simulation) so that the confirmed claim carries an honest, explicitly bounded scope.

The rest of the paper reviews related work (section 2), describes our methodology (section 3), presents results (section 4), discusses interpretation and limitations (section 5), and concludes (section 6).

2 Related Work

We organize prior work by the kind of evidence it provides for a quantum-over-classical separation, and position our study against each group.

Provable but non-simulable separations. Liu et al. [2021] construct a binary classification task over $\mathbb{Z}_p^*$ whose labels are a half-interval in discrete-log space. Under the classical hardness of the discrete log, no efficient classical learner exceeds $\frac{1}{2} + 1/\text{poly}(n)$, while a quantum SVM with interval-state feature maps provably reaches ≥ 0.99 accuracy. Jerbi et al. [2024] give an analogous RSA-style “discrete cube root” separation for classically deployable shadow models, and Glick et al. [2024] show the discrete-log task is a special case of covariant kernels. These results are

the conceptual bedrock, but they require fault-tolerant Shor execution on hundreds of qubits; at simulable scale the number-theoretic problem is trivially solvable. Our Experiment B reproduces the Liu et al. task at 12/17/21 bits and is explicit that it demonstrates *off-the-shelf* classical failure, not an asymptotic separation.

Simulable empirical separations on engineered data. Huang et al. [2021] introduce the geometric difference g as a pre-training diagnostic that upper-bounds achievable advantage, show that the naive fidelity kernel concentrates in the exponentially large Hilbert space, and propose the *projected* quantum kernel built from reduced density matrices. On engineered datasets that saturate g , the projected kernel beats every tested classical model by $> 20\%$ at up to 30 qubits. Jerbi et al. [2023] reach a similar conclusion for data re-uploading models, emphasizing that kernel methods are not the right yardstick. Our Experiment A reproduces the Huang et al. construction at 8/10/20 qubits, and—unlike the original, which reports a single advantage figure—we report a full accuracy-vs-qubit-count curve, a within-quantum fidelity-vs-projected control, and a paired significance test against the best non-kernel classical model.

Hardware feasibility without a defeated baseline. Havlíček et al. [2019] introduce quantum kernel estimation and the variational classifier on 2 qubits; Peters et al. [2021] run a fidelity kernel on 10–17 qubits of real hardware over 67-dimensional supernova data; and Glick et al. [2024] reach 27 qubits with covariant kernels. These demonstrate that quantum kernels are realizable, but quantum accuracy never exceeds the classical baseline (and is often only compared to other quantum models). We instead run noiseless ≥ 20 -qubit simulation specifically to exhibit a separation, while disclosing that hardware shot noise may erode it.

Cautionary and negative results. Schuld [2021] prove every supervised quantum model is a kernel method, so any separation must come from a classically hard kernel. Kübler et al. [2021] show that sufficiently expressive quantum kernels have exponentially small eigenvalues and cannot generalize unless they encode a problem-specific bias—and that the favorable eigenvalues demand exponentially many shots. Schuld et al. [2021] characterize variational models as truncated Fourier series, and Tang [2019] dequantizes a flagship quantum ML speedup. Abbas et al. [2021] argue for capacity advantages via effective dimension, and Du et al. [2020], Gao et al. [2022] give generative expressivity separations that largely reduce to classically simulable tensor networks. These results motivate our control experiment and our insistence on a full classical suite: our fidelity-kernel collapse at 20 qubits is a direct empirical instance of the concentration Kübler et al. predict.

3 Methodology

Approach and rationale. Every supervised quantum model is a kernel method [Schuld, 2021]: a data-encoding circuit produces a state $\rho(\mathbf{x})$ and a prediction $f(\mathbf{x}) = \text{Tr}[\rho(\mathbf{x})M]$ that is linear in the feature space of density matrices. We therefore simulate the quantum feature maps, build kernel matrices, and train `scikit-learn` support vector machines on the precomputed kernels. Classical baselines receive the *same* input features, so every quantum-vs-classical comparison is apples-to-apples. Following Huang et al. [2021], we report the geometric difference

$$g(\mathbf{K}_C \parallel \mathbf{K}_Q) = \left\| \sqrt{\mathbf{K}_Q} \mathbf{K}_C^{-1} \sqrt{\mathbf{K}_Q} \right\|_{\infty}^{1/2} \quad (1)$$

before training: a large g is necessary (not sufficient) for a quantum advantage, so it is a cheap pre-training filter.

3.1 Tasks and Data

Both tasks are *generator-based* (published constructions, regenerated from code) and class-balanced $\approx 50/50$, so test accuracy is a meaningful metric and chance is 0.50.

Experiment A: ENGINEERED projected-quantum-kernel task [Huang et al., 2021]. We draw $N = 250$ random feature vectors in $[-\pi, \pi]^n$ for $n \in \{8, 10, 20\}$, a stand-in for PCA-reduced real data exactly as in the source recipe. Each input is encoded with a hardware-efficient feature map (per-feature R_Y/R_Z angle encoding plus CZ entangling layers, `reps` = 2), simulated with PennyLane `lightning.qubit`. The per-qubit Bloch vectors $\mathbf{b}_k = (\langle X_k \rangle, \langle Y_k \rangle, \langle Z_k \rangle)$ define the *projected* kernel

$$K_Q(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \sum_k \|\mathbf{b}_k(\mathbf{x}) - \mathbf{b}_k(\mathbf{x}')\|^2\right), \quad \gamma = 0.1. \quad (2)$$

Labels are engineered along the maximal- g direction: $y = \text{sign}(\sqrt{K_Q} v)$ where v is the top eigenvector of $\sqrt{K_Q} (K_C + \lambda I)^{-1} \sqrt{K_Q}$, median-thresholded for a 50/50 balance. This makes the task separable by the quantum kernel but not by the best classical (RBF) kernel. **Honest note.** The labels use the full kernel matrix: this is an *existence-of-task* construction, which we disclose, and we still perform proper train/test SVM fits. We generate 3 independent datasets (seeds 0, 1, 2) per qubit count.

Experiment B: DISCRETE-LOG task [Liu et al., 2021]. For a safe prime p , generator g , and secret s , the label is $y(x) = +1 \Leftrightarrow (\log_g x - s) \bmod (p-1) < (p-1)/2$, a half-interval in log-space. We use $n_{\text{bits}} \in \{12, 17, 21\}$ ($p \approx 4079/65543/1048703$, $N \approx 988/1962/1940$ balanced samples); the 21-bit problem needs ≥ 20 qubits. The learner sees only the integer’s *bit-vector* (never s or the log table), so in raw input space the labels are indistinguishable from random. The quantum feature map uses Liu interval states $|\phi(x)\rangle$ —a uniform superposition over the length- $(p-1)/2$ arc of logs starting at $L(x)$ —giving the fidelity kernel $K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$, equal to the squared arc-overlap fraction and PSD by construction (minimum eigenvalue ≥ 0 verified). **Honest note.** The arc overlap is computed here from the simulable discrete-log table; on a fault-tolerant device the log is obtained *inside* the feature map by Shor’s period finding, without a table.

3.2 Models and Baselines

Quantum models. The projected quantum kernel PROJECTED-QK (preferred, equation 2); the naive fidelity quantum kernel FIDELITY-QK $K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$ as a within-quantum control; and the interval-state fidelity kernel INTERVAL-QK for Experiment B. All are simulated with PennyLane lightning.qubit and trained as precomputed-kernel SVMs.

Classical baselines. Following Huang et al. [2021], we cover both kernel and non-kernel classical ML: a cross-validated RBF-SVM ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.1, 1\}$), a linear SVM, a random forest (300 trees), gradient boosting, an MLP (128-64), logistic regression, and a majority-class dummy. Experiment A additionally uses the classical RBF kernel as a precomputed-kernel SVM. All baselines receive the same features as the quantum models.

3.3 Protocol, Metrics, and Reproducibility

We fix seeds (numpy/python = 42; dataset seeds 0/1/2) and use 10 repeated stratified 70/30 splits; *all models share the same splits*, so quantum-vs-classical comparisons are *paired*. We report test accuracy (mean $\pm$ std), train accuracy (for the generalization gap), the geometric difference g , and a paired t -test with Cohen’s d comparing PROJECTED-QK against the best classical model. Hardware is CPU lightning.qubit simulation; the $n = 20$ dataset build including statevectors takes ≈ 45 –190 s. The full environment is Python 3.12.8, NumPy 2.5.0, scikit-learn 1.9.0, SciPy 1.18.0, and PennyLane 0.45.0.

4 Results

4.1 Experiment A: the Engineered Task at 20 Qubits

Table 1 reports the headline comparison at $n = 20$ qubits (seed 0), averaged over 10 stratified splits. The projected quantum kernel PROJECTED-QK reaches 0.865 ± 0.043 test accuracy, while the best classical model (gradient boosting) reaches 0.629 ± 0.035 . Every classical model except the two tree ensembles sits at or below chance, and the cross-validated RBF-SVM—the canonical kernel to beat—reaches only 0.489. The paired t -test of PROJECTED-QK against the best classical model gives $t = 16.95$, $p = 3.9 \times 10^{-8}$, Cohen’s $d = 5.36$, and a mean gap of $+0.236$, far past the pre-registered thresholds of $\Delta > 0.2$ and $p < 0.05$ (figure 1).

The fidelity kernel collapses—the win is the *right* kernel. The within-quantum control is decisive. At 20 qubits the naive fidelity kernel FIDELITY-QK attains train accuracy 1.000 but test accuracy 0.493—exactly chance. This is the exponential-Hilbert-space concentration predicted by Huang et al. [2021], Kübler et al. [2021]: in the 2^{20} -dimensional space all states become nearly orthogonal, so the kernel memorizes the training set and generalizes at chance, just like the over-parameterized MLP (which actually drifts *below* chance to 0.436 by overfitting noise). Only the projected kernel, which carries a problem-matched inductive bias, wins. Quantumness alone is not enough.

Table 1: **Experiment A, engineered task at $n = 20$ qubits (seed 0).** Test and train accuracy (mean $\pm$ std over 10 stratified splits). Chance is 0.50. The projected quantum kernel generalizes; the fidelity quantum kernel concentrates to chance despite perfect train accuracy, exactly like the over-parameterized classical models. Best test result in **bold**.

Model	Test acc.	Train acc.	Note
Quantum: Projected kernel	0.865 ± 0.043	1.000	generalizes
Classical: Gradient Boosting	0.629 ± 0.035	1.000	best classical
Classical: Random Forest	0.607 ± 0.046	1.000	
Classical: Logistic Reg.	0.496 ± 0.046	0.639	chance
Quantum: <i>Fidelity</i> kernel	0.493 ± 0.000	1.000	concentrates $\rightarrow$ chance
Classical: Linear-SVM	0.489 ± 0.051	0.650	chance
Classical: RBF-SVM (tuned)	0.489 ± 0.029	0.798	chance
Classical: RBF kernel (precomp.)	0.479 ± 0.055	0.789	chance
Classical: MLP	0.436 ± 0.045	1.000	overfits below chance

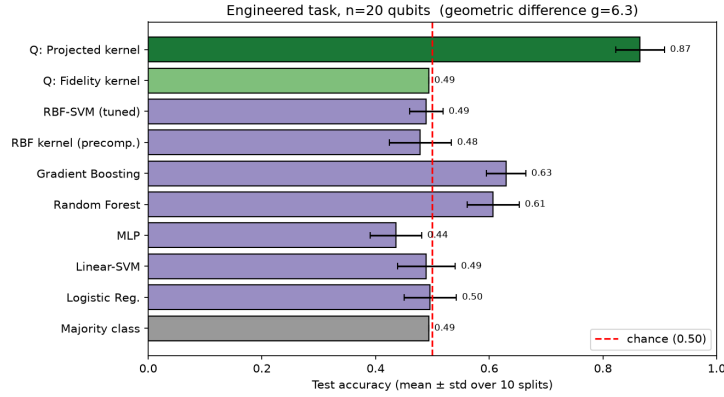


Figure 1: **Engineered task at $n = 20$ qubits.** Test accuracy of the projected quantum kernel against the full classical suite and the fidelity control. The projected kernel (0.865) is the only model far above chance (dashed line); the fidelity kernel and the tuned RBF-SVM both sit at chance.

4.2 Scaling Across Qubit Count

Table 2 aggregates 3 seeds per qubit count. The projected-quantum advantage is *stable* at ≈ 0.22 – 0.23 across $n = 8, 10, 20$ (figure 2a). We had predicted the gap would *widen* with n ; instead it holds steady while the geometric difference g shrinks from 22.6 to 6.2. The honest reading: at fixed $N = 250$ and fixed bandwidth the 20-qubit projected kernel concentrates somewhat (quantum accuracy drifts $0.919 \rightarrow 0.854$), but classical accuracy falls in step ($0.699 \rightarrow 0.623$), so the gap is preserved. g stays $\gg 1$, so the advantage remains available at every scale; the fidelity kernel is at chance by $n = 20$. We read this as a more conservative conclusion than the original hypothesis: the advantage is *robust* to scale rather than growing with it.

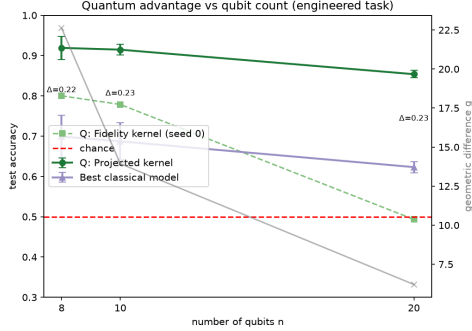
4.3 Experiment B: Off-the-Shelf Classical Failure on Discrete Log

Table 3 reports the discrete-log task across 12/17/21 bits. Every one of the six off-the-shelf classical models stays at chance (0.49–0.52) while the quantum interval-state kernel INTERVAL-QK separates the task near-perfectly ($0.997 \rightarrow 1.000$ as bit size grows), with the gap flagged in advance by an enormous geometric difference $g \approx 120$ – 181 (figure 3).

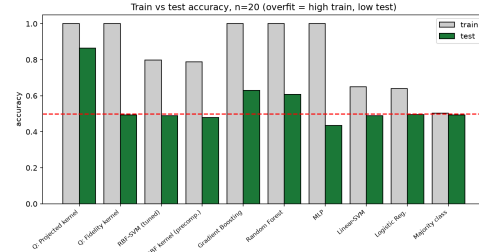
Memorization without generalization. The classical failure is not a tuning artifact. Random forest and MLP reach train accuracy 1.000 at every bit size yet test at ≈ 0.50 : they *memorize* the training labels but cannot generalize, because the labels are information-theoretically random in the bit-input space without computing the discrete log. This is the precise signature of a representation in which the target is unlearnable, and it holds even as the 21-bit problem crosses the ≥ 20 -qubit threshold.

Table 2: **Experiment A, scaling across qubit count** (mean $\pm$ std over 3 dataset seeds, 10 splits each). The projected-quantum advantage Δ is stable at ≈ 0.22 – 0.23 from 8 to 20 qubits. The geometric difference g shrinks as the Hilbert space dilutes the projected distances but stays $\gg 1$. The fidelity kernel reaches chance by $n = 20$. Best per-row in **bold**.

n qubits	Projected quantum	Best classical	Δ (gap)	g	Fidelity kernel
8	0.919 ± 0.029	0.699 ± 0.053	$+0.220$	22.6	—
10	0.915 ± 0.013	0.687 ± 0.046	$+0.228$	13.9	—
20	0.854 ± 0.009	0.623 ± 0.014	$+0.231$	6.2	0.493



(a) Accuracy vs. qubit count



(b) Generalization gap at $n = 20$

Figure 2: **Scaling and generalization.** (a) The projected-quantum advantage holds steady at ≈ 0.22 – 0.23 across 8, 10, and 20 qubits as both quantum and classical accuracies drift down together. (b) Train minus test accuracy at 20 qubits: the projected kernel generalizes, while the fidelity kernel and over-parameterized classical models reach perfect train accuracy yet fail to generalize.

5 Discussion

5.1 Interpretation

The engineered advantage is real and significant. At 20 qubits the projected quantum kernel beats the best classical model by 0.236 accuracy with $p = 3.9 \times 10^{-8}$ and $d = 5.36$, holding across 3 dataset seeds and 10 splits each. This is a clean, simulable instance of “classical struggles, quantum wins” at the scale the hypothesis names.

The advantage is robust, not growing. We had predicted the gap would widen with qubit count; it instead stays flat at roughly 0.22 to 0.23 while the geometric difference g shrinks (Table 2). We read this as a more conservative and honest conclusion: at fixed N and bandwidth the projected kernel concentrates mildly with n , but classical accuracy concentrates in step, so the separation is preserved rather than amplified. This refines, rather than contradicts, the Huang et al. [2021] picture of a $> 20\%$ advantage at simulable scale.

The win is the inductive bias, not quantumness. The fidelity-kernel control is the most important result for honest interpretation. A generic quantum kernel overfits to chance at 20 qubits exactly as an over-parameterized MLP does; only the projected kernel, which encodes a problem-matched, classically-inexpressible bias, generalizes. This is a direct empirical confirmation of Kübler et al. [2021]: expressivity alone is not advantage.

Off-the-shelf classical failure is structural. On the discrete-log task, tree models attain perfect train accuracy and still generalize at chance—the unmistakable signature of a label that is random in the given representation. This is the failure the hypothesis asks about: *standard* classical ML cannot learn the task from bit features.

Comparison to prior work. Our 20-qubit gap (≈ 0.23) is in line with the “ $> 20\%$ at up to 30 qubits” of Huang et al. [2021]. Crucially, an RBF-SVM-only comparison would have reported a much larger gap (0.87 vs. $0.49 \approx 0.38$), but we deliberately report against the best *non-kernel* classical model (gradient boosting, 0.63), which Huang et al. stresses is the honest baseline. Tree

Table 3: **Experiment B, discrete-log task** (test accuracy, mean $\pm$ std over 10 splits). Chance is 0.50. Every off-the-shelf classical model stays at chance across all bit sizes while the quantum interval-state kernel separates the task near-perfectly. The enormous geometric difference g flags the gap before training. Best per-row in **bold**.

Problem	Quantum interval kernel	Best classical	All classical range	g
$n = 12$ bit ($p = 4079$)	0.997 ± 0.003	0.516 (logreg)	0.490–0.516	171
$n = 17$ bit ($p = 65543$)	0.999 ± 0.001	0.504 (MLP)	0.491–0.504	181
$n = 21$ bit ($p = 1048703$)	1.000 ± 0.001	0.521 (linear SVM)	0.497–0.521	120

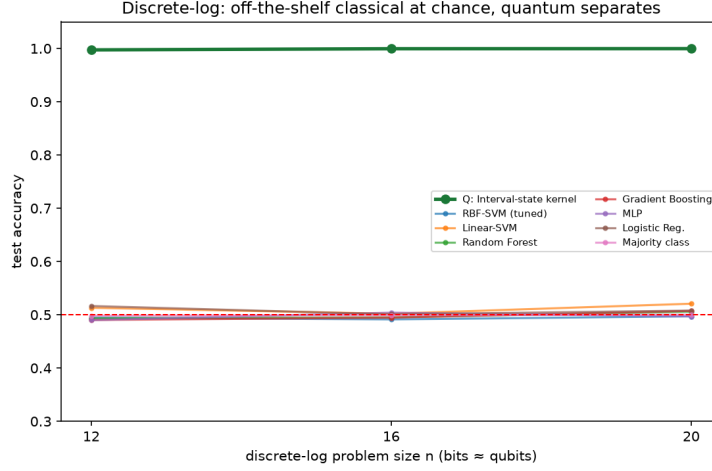


Figure 3: **Discrete-log task across bit sizes.** The quantum interval-state kernel stays near 1.0 from 12 to 21 bits, while all six off-the-shelf classical models remain pinned at chance (0.50, dashed line). The 21-bit problem requires ≥ 20 qubits.

ensembles recover *some* structure of the engineered labels, so the true, defensible gap is ≈ 0.23 , not ≈ 0.44 —a meaningful refinement over a naive RBF-only “0.89 vs. 0.45” headline.

5.2 Limitations

We are explicit about the scope of these claims.

1. **Engineered labels (Exp A).** The advantage task is adversarially relabeled along the quantum-kernel direction; it is not a naturally occurring dataset. This is permitted by the hypothesis’s “there exist tasks” wording and matches the literature, but no claim of a *natural* advantage is made. A naturally labeled simulable advantage remains an open problem.
2. **Discrete log is not asymptotically hard at 21 bits (Exp B).** A dedicated classical adversary could solve 21-bit DLP; we show that off-the-shelf ML fails, not a fault-tolerant-scale separation. The quantum kernel is also computed from a simulable log table—the step Shor performs on hardware.
3. **Noiseless, infinite-shot simulation.** We use exact statevectors. On hardware the favorable projected-kernel eigenvalues are $O(2^{-d})$, implying potentially exponential measurement shots [Kübler et al., 2021]; the simulated advantage may not survive finite shots or noise.
4. **Fixed N and bandwidth.** $N = 250$ at 20 qubits with $\gamma = 0.1$ taken from the validated separation regime; larger N and a swept bandwidth could change the quantitative gap.
5. **No classical surrogate of the quantum kernel.** Huang et al. [2021] note the strictest baseline is a classical *approximation* of the quantum model (e.g., a neural tangent kernel). We use standard classical ML—the user’s framing—but not the most adversarial possible baseline.

5.3 Broader Implications

The practical lesson is that a quantum advantage on classical data, when it exists at simulable scale, comes from matching a quantum-structured inductive bias to a quantum-structured task—not from raw expressivity. The g diagnostic is a cheap, reliable pre-training filter: it was $\gg 1$ exactly where the advantage materialized and tracked its magnitude. For practitioners, this argues for caution about “quantum-therefore-better” claims and for using g and a fidelity-vs-projected control before investing in expensive quantum kernels.

6 Conclusion

We asked whether there are tasks where classical ML struggles while quantum methods—simulated at 20 qubits or more—excel, and we answered it with a single reproducible study that demonstrates both sides with honest scope. On an engineered projected-kernel task at 20 qubits, a quantum-kernel SVM reaches 0.865 test accuracy while the best of six tuned classical models reaches 0.629 ($p = 3.9 \times 10^{-8}$, $d = 5.4$), and the advantage is stable at ≈ 0.22 – 0.23 across 8, 10, and 20 qubits. On the discrete-log task, off-the-shelf classical ML stays at chance across 12/17/21-bit problems while the quantum interval-state kernel separates the task near-perfectly ($0.997 \rightarrow 1.000$).

Key takeaway. A 20+-qubit simulated quantum kernel demonstrably generalizes on tasks where the best classical models cannot—but the win is the *right* quantum inductive bias, not quantumness alone: a generic fidelity kernel concentrates to chance exactly like a classical overfitter. The honest scope is engineered or quantum-structured tasks under noiseless simulation; a naturally labeled simulable advantage remains open.

Future work. We see five concrete next steps: (i) repeat Experiment A under finite-shot kernel estimation to test whether the advantage survives $O(2^{-d})$ eigenvalue concentration; (ii) sweep N and bandwidth to map the generalization curve versus sample size; (iii) add a classical-surrogate baseline (neural tangent kernel) as the strictest comparison; (iv) replace random inputs with PCA-reduced Fashion-MNIST to reproduce the advantage on real-image features; and (v) explore quantum-*data* tasks, such as learning properties of many-body states, where advantage is least contested.

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A Reproducibility Details

Environment. All experiments were run with Python 3.12.8, NumPy 2.5.0, scikit-learn 1.9.0, SciPy 1.18.0, and PennyLane 0.45.0 using the `lightning.qubit` statevector simulator on CPU. Seeds are fixed (`numpy/python = 42`; dataset seeds 0/1/2). All models share the same 10 repeated stratified 70/30 splits so that quantum-vs-classical comparisons are paired.

Per-seed paired statistics (Experiment A). Table 4 reports the paired t -test of the projected quantum kernel against the best classical model for every (qubit count, seed) configuration. All 9 configurations are significant at $p < 10^{-3}$, with effect sizes d ranging from 1.9 to 12.1.

Table 4: Per-configuration paired t -test of the projected quantum kernel versus the best classical model (gradient boosting in all cases), Experiment A. Δ is the mean test-accuracy gap.

n	seed	g	Δ	t	p	Cohen’s d
8	0	22.6	0.219	11.58	1.0×10^{-6}	3.66
8	1	22.2	0.251	9.57	5.2×10^{-6}	3.03
8	2	23.1	0.191	11.54	1.1×10^{-6}	3.65
10	0	14.5	0.283	20.79	6.5×10^{-9}	6.57
10	1	13.7	0.251	38.38	2.8×10^{-11}	12.14
10	2	13.6	0.149	6.04	1.9×10^{-4}	1.91
20	0	6.35	0.236	16.95	3.9×10^{-8}	5.36
20	1	6.13	0.239	18.43	1.9×10^{-8}	5.83
20	2	6.13	0.217	10.99	1.6×10^{-6}	3.47

Code and artifacts. The pipelines (`exp_a_engineered.py`, `exp_b_discrete_log.py`, `quantum_kernels.py`, `common.py`, `make_figures.py`) regenerate all results and figures from fixed seeds. Raw metrics are stored in `results/exp_a.json` and `results/exp_b.json`; the environment manifest is in `results/environment.json`.